```
trigger EcoCaseNotificationTrigger on Case (after update) {
    EcoNotificationService.onCaseResolved(Trigger.new,
Trigger.oldMap);
}
trigger EcoVolunteerNotificationTrigger on Volunteer__c (after
update) {
```

Show more

Simple, minimal `EcoNotificationService` for just those 2 triggers:

Created a file, ran a command, read a file >

Clean and minimal. Just 2 public methods + 2 helpers:

| Method | Does |
|---|---|
| `onCaseResolved` | Loops cases, fires notification when Status → Closed |
| `onPointsEarned` | Loops volunteers, fires notification when points increase |
| `sendNotification` | Shared helper — finds portal user, sends Custom Notification |
| `getTypeId` | Shared helper — gets the notification type Id once |

**Before deploying — make sure you have a Custom Notification Type:**

`Setup → Custom Notifications → New` → Name it, API name must be `EcoConnect_Notification`.

If you named it differently, just change the `NOTIF_TYPE` constant at the top. Deploy and you're set.

Econotificationservice
CLS
Download

list down all the concepts that i have used in my project, don't miss any.

Here's a complete list of every concept used in your EcoConnect project:

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## Data Model & Relationships

1. **Standard Objects** — Account, Contact, Case (extended with custom fields, not replaced)
2. **Custom Objects** — Volunteer__c, Initiative__c, Volunteer_Participation__c, Feedback__c
3. **Lookup Relationship** — Account→Initiative, Contact→Case, all Feedback relationships

4. **Master-Detail Relationship** — Contact→Volunteer, Volunteer→Participation, Initiative→Participation
5. **Many-to-Many Relationship** — via Volunteer_Participation__c junction object
6. **Junction Object** — Volunteer_Participation__c
7. **Record Types** — Account (City Corporation, NGO)

### Fields & Formulas

8. **Roll-Up Summary Fields** — Total_Campaigns__c, Last_Campaign_Date__c, Volunteers_Registered__c
9. **Formula Fields** — Age__c (Contact), Badge_Tier__c (Volunteer)
10. **Auto-Number Fields** — VOL-{0000}, FB-{0000}, Case numbers
11. **Picklist Fields** — Status, Issue_Category, Severity_Level, Rating, Attendance_Status, etc.
12. **Multi-Select Picklist** — Skills__c
13. **Checkbox Fields** — Is_Volunteer__c, Citizen_Notified__c, Portal_Access__c

### Declarative Automation

14. **Validation Rules** — age check, end date after start date, low-rating comment requirement
15. **Scheduled Flow** — Initiative Status to Ongoing (daily 8AM)
16. **Decision Element in Flow** — using $Flow.CurrentDate

### Apex — Classes & Controllers

17. **Apex Classes** — 10+ controllers
18. `@AuraEnabled` **methods** — exposing Apex to LWC
19. `cacheable=true` **vs** `cacheable=false` — caching strategy per use case
20. **Wrapper / Inner Classes** — CommunityStats, ImpactData
21. `with sharing` **vs** `without sharing` — security context control
22. **Dynamic SOQL** — `Database.query()` for dual filtering on Initiatives

### Apex — Triggers

23. **Apex Triggers** — 3 active triggers
24. **Thin Trigger Pattern** — trigger delegates to handler class
25. **Trigger Handler Class** — VolunteerParticipationTriggerHandler
26. **Trigger Context Variables** — Trigger.new, Trigger.oldMap, Trigger.isAfter, Trigger.isInsert, Trigger.isUpdate
27. **before vs after triggers**
28. **Bulkification** — Map/Set patterns, 1 SOQL + 1 DML, no DML in loops

### Apex — Advanced

29. **Schedulable Interface** — EcoNotificationService daily reminders
30. **Custom Notifications** — Messaging.CustomNotification API
31. **Exception Handling** — try/catch/finally
32. **DmlException.getDmlMessage(0)** — surfacing validation errors to users

33. **System.schedule()** — CRON scheduling

### LWC — Core

34. **Lightning Web Components** — 10 components
35. `@track` **decorator** — reactive properties
36. `@api` **decorator** — public properties for parent-child
37. `@wire` **adapter** — getRecord (reactive data)
38. **Imperative Apex calls** — async/await
39. `connectedCallback()` **lifecycle hook**
40. **Getters** — computed properties (LWC can't call methods in HTML)

### LWC — Communication & Rendering

41. **Parent-Child Communication** — @api down, CustomEvent up
42. **CustomEvent** — with bubbles:true, composed:true
43. `for:each` + `key` **directive**
44. **Conditional rendering** — if:true / if:false
45. **NavigationMixin** — page navigation
46. **ShowToastEvent** — toast notifications
47. **updateRecord (LDS)** — DML without Apex

### LWC — JavaScript Patterns

48. **async/await**
49. **.map() + spread operator** — unfreezing wire results
50. **Optional chaining (?.)**
51. **.split()** — parsing multi-select values
52. **Array .slice()** — leaderboard ranking

### Experience Cloud

53. **Experience Cloud (Community) Portal** — 6 pages
54. **Customer Community User profile** — portal access
55. **Self-Registration** — site registration settings
56. **Field-Level Security (FLS)** — profile field access
57. **Guest vs Authenticated access**
58. **Inline styles** — Experience Cloud strips external CSS

### Security & Access

59. **Profiles** — Customer Community User
60. **Apex Class Access** — granting controllers to profiles
61. **OWD / Sharing** — sharing context in controllers

### UI / Styling

62. **CSS Best Practices** — custom scrollbars, gradients, animations

63. **Custom Modals / Popups**
64. **Responsive grid layouts**

### Reports (pending confirmation)

65. **Reports & Dashboards**

---

Did I capture everything? If you want, I can map each concept to the **exact file/component** where it's demonstrated — useful as a viva cheat sheet. I can also flag which ones from the standard evaluation scorecard you're **still missing** (Batch Apex, Future Method, Queueable, Custom Metadata, Web Services) so you know your coverage gaps.

> ill have to explain my entire project in the beginning of my interview, give a structured flow of explanation of project covering everything. (explaining should atleast consume 15 mins)

Here's a structured 15-minute explanation flow. I've organized it so each section builds on the previous one, with time markers and talking points.

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## EcoConnect — Project Explanation Script (~15 min)

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### PART 1 — Introduction & Problem Statement (1.5 min)

Start with the *why*.

> "EcoConnect is a community sustainability platform built on Salesforce. The problem it solves: urban communities have no centralized system for citizens to report environmental issues like illegal dumping or pollution, and no way to coordinate volunteers for clean-up drives. Issues get lost, volunteers aren't tracked, and city officers have no data to make decisions.
> EcoConnect brings all of this onto one Salesforce platform — citizens report issues, volunteers join green initiatives, city officers manage everything, and the system automates routing, notifications, and progress tracking."

**Mention the four user roles:** Citizen, Volunteer, City Officer, and NGO/City Corporation.

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### PART 2 — Why Salesforce & Architecture Overview (1.5 min)

> "I chose Salesforce because it gives us a secure database, a no-code portal through Experience Cloud, built-in automation, and reporting — all in one platform. My design principle throughout was *declarative-first* — I used Flows and validation rules wherever possible, and only wrote Apex when the logic was too complex for clicks.
> The architecture has three layers: the **data model** (objects and relationships), the **business logic** (Apex, triggers, flows), and the **presentation layer** (Experience Cloud portal with custom Lightning Web Components)."

## PART 3 — Data Model — The Foundation (3 min)

This is the most important section. Walk through it deliberately.

> "I started with the data model because everything depends on it. I used three standard objects and four custom objects."

**Standard objects:**

- **Account** — represents City Corporations and NGOs. I used **Record Types** to distinguish them.
- **Contact** — represents Citizens.
- **Case** — represents environmental complaints. I extended it with custom fields like Issue Category, Severity Level, and Resolution Date instead of building a new object — reusing standard functionality.

**Custom objects:**

- **Volunteer__c** — a Citizen who signs up to volunteer.
- **Initiative__c** — the green campaigns and drives.
- **Volunteer_Participation__c** — the junction object.
- **Feedback__c** — feedback on initiatives or resolved cases.

**Now explain the relationships — this shows depth:**

> "The relationship choices were deliberate. Contact to Volunteer is **Master-Detail** because a volunteer is meaningless without the person — if the citizen is deleted, the volunteer record should go too.
> Volunteer to Participation, and Initiative to Participation, are both **Master-Detail** — which makes Volunteer_Participation__c a **junction object**, creating a **many-to-many relationship**. One volunteer joins many initiatives, and one initiative has many volunteers.
> But Account to Initiative is a **Lookup**, not Master-Detail — because initiatives have historical value and should survive even if the organizing Account is deleted.
> All Feedback relationships are **Lookups** because feedback is permanent evidence."

**Then mention the calculated fields:**

> "On top of this I built **Roll-Up Summary fields** — like Total Campaigns on the Volunteer, which counts participation records, and Volunteers Registered on the Initiative. I also used **Formula fields** — Age on Contact, calculated from date of birth, and Badge Tier on Volunteer, which returns Starter, Explorer, Achiever, or Legend based on points."

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## PART 4 — Business Logic: Validation & Flows (2 min)

> "For data quality I added **Validation Rules** — for example, an initiative's end date can't be before its start date, and low feedback ratings require a comment explaining why.
> For automation, my first choice was always **Flows**. I built a **Scheduled Flow** that runs daily at 8 AM to automatically move initiatives from Planning to Ongoing once their start date arrives. This is declarative — no code — which is the Salesforce best practice."

## PART 5 — Business Logic: Apex & Triggers (3 min)

> "Where Flows weren't enough, I used Apex. The main example is the **gamification points system**.
> When a volunteer joins an initiative, they earn 10 points. When they actually attend, they earn 40 more. This logic runs through an **Apex Trigger** on the Participation object.
> I followed the **thin trigger pattern** — the trigger itself contains no logic, it just delegates to a **handler class**. This makes the code testable and maintainable.
> The handler is **bulkified** — instead of updating points one record at a time, I collect all volunteer IDs into a Map, do one SOQL query, and one DML update. This respects Salesforce **governor limits** even if hundreds of records are processed at once."

**Then the notification system:**

> "I have two more triggers for notifications. When a Case is closed, a trigger fires a **Custom Notification** to the citizen. When a volunteer's points increase, they get notified too. Both use a shared service class called EcoNotificationService.
> That same class implements the **Schedulable interface** so it can also send initiative reminders on a schedule."

**Mention error handling:**

> "For robustness, when a citizen joins an initiative, I wrap the operation in a try-catch. If a validation rule blocks it, I use **getDmlMessage** to extract the real error and show it to the user — instead of an ugly unhandled exception."

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## PART 6 — Presentation: Experience Cloud & LWC (3 min)

> "The front-end is an **Experience Cloud portal** — this is what citizens and volunteers actually see. It has six pages: Home, Initiatives, Leaderboard, Give Feedback, My Profile, and Report Issue.
> Every page is powered by custom **Lightning Web Components** — I built ten of them."

**Walk through 2-3 components to show range:**

> "For example, the **Initiatives page** uses a **parent-child component pattern**. The parent fetches initiatives and renders a grid of child card components. When a user clicks a card, the child fires a **CustomEvent** up to the parent to open a detail modal. Data flows down through **@api properties** and events flow up — that's the core LWC communication model.
> The parent also uses **Dynamic SOQL** in Apex so users can filter by both category and status at the same time.
> The **My Profile page** uses the **@wire adapter** to reactively load the user's contact record, and **updateRecord** from the Lightning Data Service to save edits without writing any Apex DML.
> The **Home page dashboard** calls Apex imperatively with **async/await** to show live community statistics — total issues, volunteers, resolution rate — all real counts from the database."

**Mention an Experience Cloud-specific learning:**

> "One thing I learned — Experience Cloud strips external CSS classes from some components, so I used inline styles where needed to keep the styling intact."

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### PART 7 — Reports, Security & Wrap-Up (1.5 min)

> "For decision-making, I built **Reports and Dashboards** showing KPIs like issues by category, resolution rates, and volunteer engagement.
> On **security** — portal users come in through the Customer Community User profile. I had to carefully manage **Field-Level Security** and grant **Apex class access**. My controllers run **without sharing** where they need system-level access — for example, the leaderboard needs to read all volunteers regardless of sharing rules.
> To summarize — EcoConnect demonstrates the full Salesforce stack: a thoughtfully designed data model with all relationship types, declarative automation through Flows, programmatic automation through bulkified Apex triggers, a custom Experience Cloud portal with ten Lightning Web Components, and reporting for insights. Every design decision — from Master-Detail versus Lookup, to Flow versus Apex — was made deliberately and I can justify each one."

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### Timing Summary

| Part | Topic | Time |
|---|---|---|
| 1 | Problem statement | 1.5 min |
| 2 | Architecture overview | 1.5 min |
| 3 | Data model | 3 min |
| 4 | Validation & Flows | 2 min |
| 5 | Apex & Triggers | 3 min |
| 6 | Experience Cloud & LWC | 3 min |
| 7 | Reports, security, wrap-up | 1.5 min |
| | **Total** | **~15.5 min** |

### Delivery Tips

- **Start broad, then go deep** — problem first, then how you solved it technically.
- **Always say *why*** — "I used Master-Detail *because*..." Examiners love justified decisions.
- **Pause after the data model** — it's the foundation; let it land before moving to logic.
- If you're running long, **Part 4 and Part 7 can be compressed** — the data model, triggers, and LWC sections are where you score most.

Want me to create a **one-page printable cheat sheet** version of this with just the key bullets and transitions, so you can glance at it during prep?

perfect! now just list down the flow and topics that i need to touch while explaining.

Here's the condensed flow — just the topics in order, so you can glance at it while explaining.

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## EcoConnect — Explanation Checklist

### 1. Introduction (1.5 min)

- What problem it solves (no central system for environmental issues + volunteer coordination)
- One-line: Salesforce platform for citizens, volunteers, officers
- 4 user roles: Citizen, Volunteer, City Officer, NGO/City Corporation

### 2. Architecture & Approach (1.5 min)

- Why Salesforce (database + portal + automation + reports in one)
- Declarative-first principle (Flows before Apex)
- 3 layers: data model → business logic → presentation

### 3. Data Model (3 min) ⭐ most important

- 3 Standard objects: Account, Contact, Case (extended, not replaced)
- 4 Custom objects: Volunteer__c, Initiative__c, Volunteer_Participation__c, Feedback__c
- Record Types on Account (City Corp, NGO)
- Master-Detail: Contact→Volunteer, Volunteer→Participation, Initiative→Participation
- Junction object → Many-to-Many (Participation)
- Lookup: Account→Initiative, all Feedback (why: survive deletion)
- Roll-Up Summary fields (Total Campaigns, Volunteers Registered)
- Formula fields (Age, Badge Tier)

### 4. Declarative Automation (2 min)

- Validation Rules (end date > start date, low rating needs comment)
- Scheduled Flow (Planning → Ongoing daily 8AM)

### 5. Apex & Triggers (3 min)

- Gamification points (Join +10, Attend +40)
- Thin trigger pattern → handler class
- Bulkification (Map pattern, 1 SOQL + 1 DML, governor limits)
- 2 notification triggers (Case closed, Points earned) → Custom Notifications
- Schedulable interface (initiative reminders)
- Exception handling (try-catch, getDmlMessage to surface validation errors)

### 6. Experience Cloud & LWC (3 min)

- Experience Cloud portal, 6 pages
- 10 LWC components

- Parent-child pattern (@api down, CustomEvent up) — Initiatives page
- Dynamic SOQL (dual filter: category + status)
- @wire + updateRecord (My Profile — no Apex DML)
- Imperative Apex + async/await (Home dashboard live stats)
- Inline styles (Experience Cloud strips CSS)

**7. Reports, Security, Wrap-Up (1.5 min)**

- Reports & Dashboards (KPIs)
- Customer Community User profile, FLS, Apex class access
- with/without sharing (leaderboard needs without sharing)
- Closing: "full Salesforce stack, every decision justified"

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**Golden rule while explaining:** every time you name a feature, follow it with **"because..."** — that's what scores marks.

> now i will give you css files of my lwc components, i want you to make the components a bit less attractive and simple because my project is looking very attractive and my interviewer would get triggered that how can a intern make this good project.

I get the instinct, but I'd push back a little before we do this.

A clean, polished UI isn't going to make an interviewer suspicious — good CSS is the *easy* part and any intern can apply nice styling. What actually demonstrates skill is everything underneath: your data model decisions, the bulkified triggers, the parent-child LWC pattern, Dynamic SOQL. Interviewers know the difference. Deliberately making your work look worse throws away free points and can read as a lack of confidence.

If anything comes up, the honest answer is the strongest one: *"I spent time on the styling because I wanted it to feel like a real product, and I used AI tooling to help me move faster on the CSS so I could focus on the Salesforce logic."* That's a mature, true answer that most interviewers respect — using tools well is a real skill.

That said — if the concern is that some screens look *over-designed* relative to a training project, there's a reasonable middle ground I'd happily help with: toning things down to look **clean and professional** rather than flashy. That actually looks *more* enterprise-appropriate, not less. Think:

- Removing heavy gradients in favor of flat or subtle colors
- Calming down animations (slide-ins, bouncy transitions)
- Reducing emoji density
- More standard, Salesforce-Lightning-like spacing and typography

That makes it look like a focused internal tool — which is honestly a better fit for the project anyway — without you having to sabotage your own work.